

# A Focus-Cost Model for Meeting Placement

## 1 Motivation

Meeting fatigue is usually attributed to the number of meetings. This document sets out a model in which the dominant factor is instead *fragmentation*: where meetings sit within a day, rather than how much of the day they occupy.

Two hours of meetings placed at 09:00 and 15:00 leave nobody with more than three hours of uninterrupted time. The same two hours booked 09:00 to 11:00 leave a five hour block. The load is identical; the days are not.

## 2 Notation

All quantities are defined for a single person on a single day. The day is the unit of computation because the interactions that matter, namely fragmentation, fatigue and the loss of a lunch break, occur within a day and not across days.

### Sets

---

|                  |                                                    |
|------------------|----------------------------------------------------|
| $W = [w_s, w_e]$ | working window                                     |
| $M$              | the person's meetings on the day                   |
| $G(M)$           | gaps: free intervals within $W$ induced by $M$     |
| $R(M)$           | runs: maximal contiguous stretches of meeting time |
| $A(m)$           | attendee list of a candidate meeting $m$           |
| $N, n =  N $     | meetings in a person-day, used for attribution     |

---

Gaps and runs are derived from  $M$ , not stored. Overlapping meetings merge into a single run.

### Parameters

---

| Symbol    | Meaning               | Default     | Kind        |
|-----------|-----------------------|-------------|-------------|
| $c$       | re-entry cost         | 20 min      | measurement |
| $\lambda$ | fatigue rate          | 0.4         | measurement |
| $r^*$     | fatigue threshold     | 90 min      | measurement |
| $\mu$     | lunch penalty         | 60 min      | policy      |
| $L$       | lunch window          | 12:00–14:00 | policy      |
| $\ell$    | required lunch length | 60 min      | policy      |

---

Measurement parameters are intended to be fitted against self-reported data and are not asserted. Policy parameters encode a norm the organisation chooses and are not subject to calibration.  $\lambda$  is dimensionless; all other parameters are in minutes.

### 3 The model

#### 3.1 Gap yield

A free interval is not worth its face value, since some minutes at the start of any free stretch are lost re-entering the task:

$$u(g) = \max(0, |g| - c)$$

A 60-minute gap yields 40 minutes of usable focus. A 15-minute gap yields nothing at all. The re-entry cost is paid once per gap, so a day broken into four pieces pays it four times.

#### 3.2 Fatigue

Long unbroken runs of meetings degrade what follows them:

$$\Phi(M) = \lambda \sum_{r \in R(M)} \max(0, |r| - r^*)$$

This is convex in run length, so the fourth consecutive meeting is charged more than the first.

#### 3.3 Lunch protection

A day leaving no adequate break within the lunch window is penalised. The check is against raw gap lengths rather than gap yields, and is all or nothing:

$$\Lambda(M) = \mu \cdot \mathbf{1} \left[ \nexists g \in G(M) : g \subseteq L \wedge |g| \geq \ell \right]$$

#### 3.4 Usable focus time and the day penalty

Write  $Y(M)$  for the total yield of the day's free intervals:

$$Y(M) = \sum_{g \in G(M)} \max(0, |g| - c)$$

Usable focus time subtracts the two charges from that yield:

$$U(M) = Y(M) - \Phi(M) - \Lambda(M)$$

An empty day contains a single gap spanning  $W$ , so

$$U(\emptyset) = |W| - c$$

and the day penalty is the shortfall against it:

$$P(M) = U(\emptyset) - U(M)$$

$P$  is denominated in minutes of focus time destroyed. An empty day scores zero; a fully booked day scores  $|W| - c$ .

### 3.5 Collected form

Expanding every term gives the penalty function in full, as three contributions to the shortfall against a clear day:

$$P(M) = (|W| - c) + \underbrace{\sum_{g \in G(M)} -\max(0, |g| - c)}_{\text{fragmentation}} + \underbrace{\lambda \sum_{r \in R(M)} \max(0, |r| - r^*)}_{\text{fatigue}} + \underbrace{\mu \cdot \mathbf{1}[\nexists g \in G(M) : g \subseteq L \wedge |g| \geq \ell]}_{\text{lunch}}$$

The fragmentation term is the negative of the day's focus yield. The more usable focus the day retains, the smaller the shortfall, so a day broken into more pieces contributes more to  $P$  through paying  $c$  once per gap.

## 4 Marginal cost

The cost of a candidate meeting is the increase in day penalty it causes, summed over everyone invited:

$$\text{Cost}_p(m \mid M) = P(M \cup \{m\}) - P(M) \quad \text{Cost}(m) = \sum_{p \in A(m)} \text{Cost}_p(m \mid M_p)$$

Since  $U(\emptyset)$  cancels between the two evaluations, this decomposes into the change in each component:

$$\text{Cost}_p(m \mid M) = \underbrace{[Y(M) - Y(M \cup \{m\})]}_{\text{focus yield lost}} + \underbrace{[\Phi(M \cup \{m\}) - \Phi(M)]}_{\text{fatigue added}} + \underbrace{[\Lambda(M \cup \{m\}) - \Lambda(M)]}_{\text{lunch added}}$$

Marginal cost is a function of calendar state, not a property of the meeting. It is not additive: subadditive when meetings abut, since the second pays only its duration, and superadditive once a combined run exceeds  $r^*$ .

## 5 Closed form under the fragmentation term

The results in this section hold when  $\Phi$  and  $\Lambda$  do not bind. Both additional terms deliberately break them.

For a candidate of duration  $D$  placed inside a single gap of length  $g$ , leaving fragments  $a$  on the left and  $b$  on the right with  $a + D + b = g$ :

$$\text{Cost} = \max(0, g - c) - \max(0, a - c) - \max(0, b - c) = \begin{cases} D + c, & a \geq c \wedge b \geq c \\ D + a, & a < c \wedge b \geq c \\ D + b, & a \geq c \wedge b < c \\ g - c, & a < c \wedge b < c \end{cases}$$

Cost is therefore constant at  $D + c$  across the interior of a gap and falls linearly to  $D$  at its edges. The fourth case gives a cost below the meeting's own duration, which is correct: a candidate filling a gap entirely consumes time that was partly lost to re-entry regardless.

## 5.1 Spread

It follows that when a flush position exists for every attendee, the difference between the dearest and cheapest bookable slot is

$$\max_s \text{Cost}(m_s) - \min_s \text{Cost}(m_s) = |A| \cdot c$$

independent of duration and of the shape of the day. Each attendee pays either  $D$  or  $D + c$ , so the difference is  $c$  per attendee.

This is a prediction rather than an observation, and it was confirmed on 13 of 20 randomly generated scenarios. The remaining seven lacked a candidate position leaving every attendee flush, and so never reached the lower bound.

## 6 Retrospective attribution

Marginal cost depends on the order in which meetings were booked, which is invisible on a calendar, so it cannot be used to rank meetings against one another. For a person-day with meeting set  $N$ , allocate the day penalty by Shapley value: each meeting's marginal contribution, averaged over every order in which the meetings could have arrived.

$$\varphi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [P(S \cup \{i\}) - P(S)]$$

This is the unique allocation satisfying efficiency, symmetry, the null player property and additivity. Efficiency,

$$\sum_{i \in N} \varphi_i = P(N),$$

is the property that matters in practice: attributions across a day sum exactly to that day's penalty, so per-meeting figures reconcile with per-person totals.

Exact computation requires  $n \cdot 2^{n-1}$  evaluations of  $P$ , which is negligible for the meeting counts found in a single day. Above roughly  $n = 12$  the implementation samples permutations by Monte Carlo and reports a standard error.

### 6.1 Which measure belongs where

Marginal cost is prospective and answers *what will this booking cost*, so it belongs in the scheduling view, where state-dependence is information rather than distortion. Shapley attribution is retrospective and answers *what share of this day's damage does this meeting own*, so it belongs on the calendar, where the figures are expected to sum.

## 7 Limitations

**Linear gap yield.**  $u(g)$  treats a minute of free time as equally valuable regardless of the block it sits in, so breaking a four-hour block and breaking a forty-minute gap are priced similarly. Weighting yield by block quality, for instance  $u(g) = (g - c) \min(1, g/g^*)$ , would distinguish them at the cost of one further parameter and of the closed form above.

**No benefit term.** The model prices cost and never value. Its unconstrained optimum is to hold no meetings, which is plainly wrong. A sampled post-meeting value signal would give a cost-versus-value view, but requires participation in a way the cost side does not.

**Uncalibrated parameters.** The measurement parameters are starting values. The intended fitting procedure is a grid search maximising Spearman rank correlation between  $P$  and participants' own rankings of their past weeks.